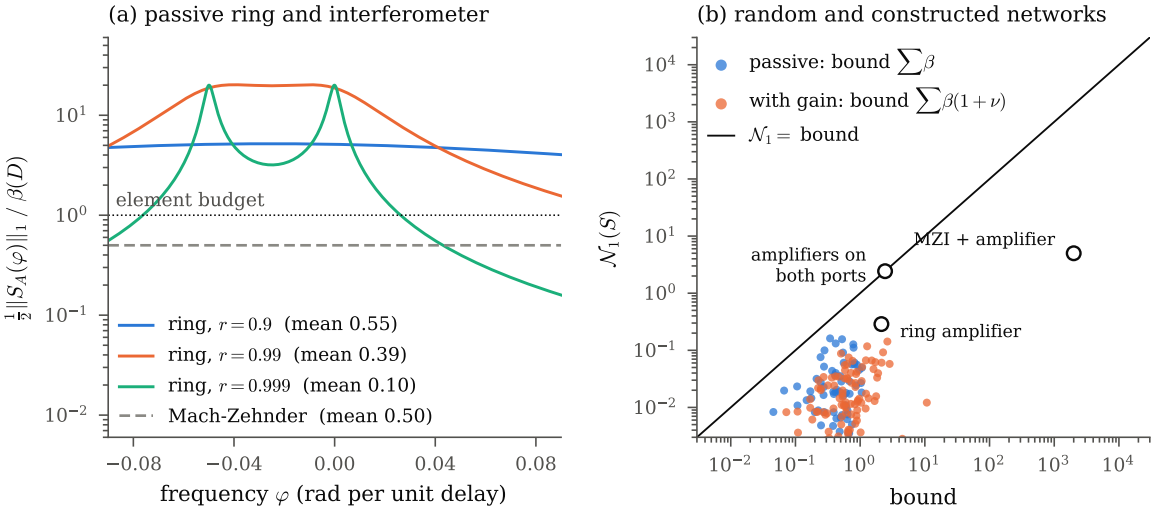
(a) passive ring and interferometer
$\frac{1}{2}\|S_A(\varphi)\|_1 \;/\; \beta(D)$
element budget
ring, $r = 0.9$ (mean 0.55)
ring, $r = 0.99$ (mean 0.39)
ring, $r = 0.999$ (mean 0.10)
Mach-Zehnder (mean 0.50)
frequency φ (rad per unit delay)
(b) random and constructed networks
passive: bound $\sum\beta$
with gain: bound $\sum\beta(1+\nu)$
$\mathcal{N}_1 =$ bound
$\mathcal{N}_1(S)$
amplifiers on both ports
MZI + amplifier
ring amplifier
bound